Temporal Analysis of Heart Disease and Stroke Mortality Trends Across the United States (1999–2019)

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***Abstract*—This paper presents a comprehensive temporal data analysis of the U.S. Heart Disease and Stroke Mortality dataset [1], which records age-adjusted mortality rates across U.S. counties and states from 1999 to 2019. Approximately 5.77 million records were preprocessed into a national-level annual time series of 21 observations. Temporal patterns were investigated through classical additive decomposition, STL decomposition (trend strength $F_t = 0.9962$, seasonal strength $F_s = 0.0000$), moving-average detrending, first-order differencing, Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) stationarity tests, Ljung-Box serial correlation tests, ACF/PACF analysis, and temporal regime classification. Regime analysis found 19 of 20 years in persistent HH or LL regimes, confirming extreme temporal clustering. Six forecasting models — naïve Random Walk, Linear Regression, Auto-ARIMA (selected as ARIMA(1,1,2), AIC = 117.87), Holt's Linear Trend, Holt-Winters Additive ($\gamma = 0.0000$), and Auto-SARIMA — were validated on a held-out test set (2015–2019, 5 observations). Auto-ARIMA(1,1,2) achieved the best holdout performance (RMSE = 4.04, MAPE = 0.79%). Residual diagnostics confirmed white-noise residuals (Ljung-Box $p > 0.98$ at all lags), formally validating model adequacy.**

***Index Terms*—Temporal Data Analytics, Heart Disease Mortality, ARIMA, Stationarity, ADF, KPSS, Ljung-Box, ACF, PACF, Exponential Smoothing, Decomposition, MAPE, Temporal Regimes, Residual Diagnostics**

I. Dataset Description

A. Source and Overview

The Heart Disease and Stroke Mortality dataset was obtained from a publicly available repository on Kaggle [1]. It contains U.S. county- and state-level age-adjusted mortality rate data for heart disease and stroke, collected from 1999 through 2019. The dataset contains approximately 5.77 million rows across 21 columns. Table I summarises the key structural properties of the dataset used throughout this study.

B. Dataset Structure

Each row represents a single mortality observation with the following key attribute groups:

Temporal: The `Year` column records the observation year (1999–2019), originally stored as an object type requiring numeric conversion.

TABLE I
Summary Statistics of the Heart Disease Mortality Dataset

Property	Value
Total rows	5,770,240
Columns	21
Records with valid `Data_Value`	3,404,765
Mean `Data_Value`	448.88
Std `Data_Value`	625.79
Temporal span	1999–2019 (21 years)
Geographic levels	State and County
Demographic strata	Age groups, sex, race/ethnicity

Geographic: `LocationAbbr`, `LocationDesc`, `GeographicLevel`, and `LocationID` identify the U.S. state or county.

Mortality Measurement: `Data_Value` stores the age-adjusted mortality rate (deaths per 100,000 population); `Confidence_limit_Low` and `Confidence_limit_High` provide 95% confidence intervals.

Stratification: `Stratification1`, `Stratification2`, and `Stratification3` encode demographic breakdowns including age groups (e.g., *Ages 35–64 years*, *Ages 65 years and older*), sex, and race/ethnicity.

Topic and Class: `Topic` distinguishes heart disease from stroke; `Class` indicates the broader disease category.

C. Motivation for Dataset Change

The team's original Assignment 1 dataset (TreeSEU) was a purely spatial point-pattern dataset with no temporal dimension. Since Assignment 2 focuses on temporal analytics, it was unsuitable. The Heart Disease and Stroke Mortality dataset was selected because it spans over two decades of annually recorded observations, providing a well-defined time axis.

D. Characteristics Making the Dataset Suitable

Five characteristics make this dataset suitable for temporal analysis. First, the `Year` column provides a regularly sampled temporal index. Second, the 21-year span provides sufficient data for decomposition, stationarity testing, and forecasting. Third, continuously measured mortality rate values

TABLE II
METHODS APPLIED AND THEIR DATASET-SPECIFIC PURPOSE

Method	Purpose / Dataset Motivation
STL / Classical Decomposition	Quantify F_t, F_s; confirm trend-only structure
MA Detrend + Differencing	Remove non-constant mean; establish $d = 1$
ADF + KPSS	Dual stationarity verdict; resolve ADF ambiguity on $n = 21$
Ljung-Box	Confirm serial correlation; validate white-noise residuals
ACF / PACF	Guide ARIMA order ($p = 1$, $q = 1$–2)
Temporal Regime Classification	Detect persistent HH/LL phases; corroborate ACF
ARIMA (AIC grid)	Primary forecasting; $d = 1$ confirmed by stationarity
Holt's / Holt-Winters	Trend model for $F_s = 0$ series; seasonal baseline
Random Walk + OLS	Lower-bound baselines; quantify value of AR modelling

are compatible with ARIMA [2] and Exponential Smoothing [3]. Fourth, stratification attributes support disaggregated sub-series analysis. Fifth, the nationwide scope enables both aggregate and location-specific temporal analyses.

II. APPLICABLE TEMPORAL STATISTICAL ANALYSIS TECHNIQUES

Table II summarises how each method maps to a specific dataset characteristic. The paragraphs below give the dataset-specific justification for each choice.

Decomposition: The 21-year span and monotonic decline make the series amenable to additive decomposition [3] [4]. STL strength metrics ($F_s = 0.0000$, $F_t = 0.9962$) formally confirm trend dominance and directly rule out seasonal forecasting models.

Detrending and Differencing: The downward trend in Figure 1 violates ARIMA stationarity. First-order differencing suffices because the trend is approximately linear.

Stationarity (ADF + KPSS): ADF alone is unreliable on 21 points with a deterministic trend — here it nominally accepts stationarity at $p = 0.018$ while KPSS correctly rejects it. Using both tests exposes this disagreement and motivates the three-source corroboration in Section IV-C.

Ljung-Box: Serial correlation is expected in a mortality series driven by multi-year policy cycles. Ljung-Box [5] yields $Q = 49.03$ ($p < 0.0001$) at lag 6 on the original and $p > 0.98$ on model residuals — serving both as a pre-modelling diagnostic and a post-fit adequacy check.

ACF/PACF: The PACF spike at lag 1 and gradual ACF decay confirm AR(1) structure and directly motivate ARIMA(1,1,1) and ARIMA(1,1,2) as primary candidates.

Temporal Regime Classification: 19 of 20 years fall in persistent HH or LL regimes, independently corroborating the autocorrelation evidence.

ARIMA: Confirmed autocorrelation and $d = 1$ sufficiency make ARIMA the natural primary model [2]. AIC search over 18 configurations selects ARIMA(1,1,2).

Holt's / Holt-Winters: $F_s = 0.0000$ is precisely the scenario Holt's linear trend model is designed for [3]. Holt-Winters confirms the prediction: optimised $\gamma = 0.0000$.

Baselines: Random Walk and OLS set a quality floor and quantify the value of autocorrelation modelling.

III. STATISTICAL ANALYSIS TECHNIQUES

A. Augmented Dickey-Fuller (ADF) Test

The ADF test [6] evaluates the null hypothesis H_0 that a unit root is present (non-stationary). The test statistic is:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{k} \delta_i \Delta y_{t-i} + \varepsilon_t \tag{1}$$

where $\gamma < 0$ indicates stationarity. Rejection at $p < 0.05$ implies the series is stationary.

B. KPSS Test

The KPSS test [7] reverses the hypothesis: H_0 is that the series is stationary. Rejection ($p < 0.05$) indicates non-stationarity. Using ADF and KPSS together produces a more robust verdict: if ADF rejects and KPSS fails to reject, the series is confirmed stationary; if both reject or disagree, the result is inconclusive and requires further investigation.

C. Ljung-Box Test

The Ljung-Box statistic [5] tests whether the first h autocorrelations of a series are jointly zero:

$$Q(h) = n(n+2) \sum_{k=1}^{h} \frac{\hat{\rho}_k^2}{n-k} \tag{2}$$

Rejection of H_0 ($p < 0.05$) confirms that temporal structure remains and must be explicitly modelled.

D. STL Component Strength

Following Hyndman and Athanasopoulos [3], seasonal strength F_s and trend strength F_t are defined as:

$$F_s = \max\left(0,\ 1 - \frac{\mathrm{Var}(R_t)}{\mathrm{Var}(S_t + R_t)}\right) \tag{3}$$

$$F_t = \max\left(0,\ 1 - \frac{\mathrm{Var}(R_t)}{\mathrm{Var}(T_t + R_t)}\right) \tag{4}$$

where T_t, S_t, and R_t are the trend, seasonal, and residual components respectively. Values above 0.64 indicate strong components.

E. ARIMA Model

An ARIMA(p, d, q) model [2] combines autoregression (AR), differencing (I), and moving average (MA) components:

$$\phi(B)(1-B)^d y_t = \theta(B)\varepsilon_t \tag{5}$$

where $\phi(B)$ and $\theta(B)$ are polynomials in the backshift operator B, and ε_t is white noise. Model order selection is guided by ACF/PACF plots and AIC grid search over (p, d, q) combinations. The AIC-selected model in this study is ARIMA(1,1,2) with AIC = 117.87.

TABLE III
SELECTED ANNUAL MEAN MORTALITY RATE VALUES (NATIONAL AGGREGATE)

Year	Mean Mortality Rate (per 100k)
1999	654.38
2000	637.86
2003	589.62
2006	508.71
2010	447.86
2014	415.25
2019	$\approx$406

F. Holt's Exponential Smoothing

Holt's linear trend model [3] updates level L_t and slope T_t recursively:

$$L_t = \alpha y_t + (1-\alpha)(L_{t-1} + T_{t-1}) \tag{6}$$

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1} \tag{7}$$

The h-step-ahead forecast is $\hat{y}_{t+h} = L_t + h \cdot T_t$. Optimised parameters on this dataset: $\alpha = \beta = 0.6638$ for Holt's Linear, and $\alpha = \beta = 0.6649$, $\gamma = 0.0000$ for Holt-Winters Additive.

G. Model Evaluation Metrics

Holdout performance is evaluated using RMSE and MAPE [8]:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{8}$$

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{9}$$

MAPE expresses forecast error as a percentage of the actual mortality rate, making results interpretable independently of the absolute scale of the series.

IV. EXPERIMENTS

A. Preprocessing and Time Series Construction

The dataset was loaded using `pandas` [8]. The `Year` column, stored as an object type, was converted to a numeric integer. Rows with missing or non-parseable year or mortality values were dropped, reducing the dataset from 5,770,240 rows to 3,404,765 valid records. The dataset contains two stratification strata: *Ages 35–64 years* and *Ages 65 years and older* — no `Overall` stratum was present in this dataset version, so the full filtered dataset (both age groups combined) was used to construct the national aggregate. The year index was converted to a `DatetimeIndex` and the annual year-start frequency (`YS`) was assigned.

A national-level time series was constructed by computing the mean `Data_Value` grouped by `Year`, yielding 21 annual observations (1999–2019). The `iloc`-based train-test split was preferred over `loc`-based slicing, which produced an empty test set due to `DatetimeIndex` boundary behaviour. The 80/20 split yields **training: 1999–2014 (16 observations)** and **test: 2015–2019 (5 observations)**. Table III shows selected values from the constructed time series.

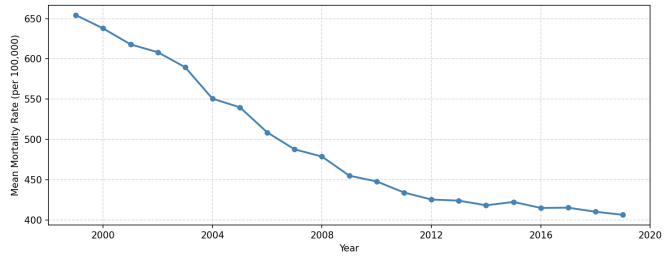

Fig. 1. National annual mean heart disease and stroke mortality rate (1999–2019). The series declines monotonically from $\approx$654 to $\approx$406 deaths per 100,000 — a 38% reduction over 20 years. The absence of oscillation confirms a trend-dominated series with negligible seasonal variation, motivating non-seasonal ARIMA models.

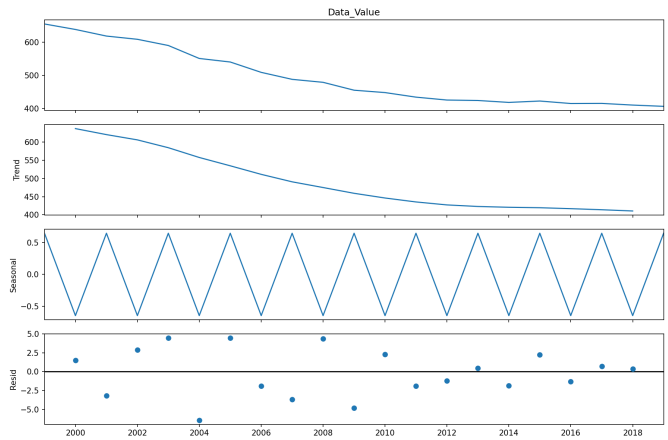

Fig. 2. Classical additive decomposition (period = 2). The trend panel reveals a steady long-term decline as the dominant structure. The seasonal component is negligibly small, consistent with annually sampled data. The residual panel fluctuates unsystematically, confirming model adequacy.

B. Decomposition and Detrending

1) Classical Additive Decomposition: Classical additive decomposition [3] was applied with `period=2`. Figure 2 shows all four components. The trend component dominates; the seasonal component is negligibly small; and the residual fluctuates randomly around zero.

2) STL Decomposition: STL [4] was applied with `robust=True`, `period=2`. Figure 3 shows the four panels. The formal component strengths (Equation 4) are $F_s = 0.0000$ and $F_t = 0.9962$. These values directly predict that seasonal models will offer no advantage — confirmed in Section V by Holt-Winters optimising to $\gamma = 0.0000$.

3) Detrending and Differencing: Figure 4 compares two detrending approaches. Panel (b): MA-detrended series, ADF statistic $= -2.1986$, $p = 0.2068$ (non-stationary). Panel (c): first-differenced series, ADF statistic $= -9.7974$, $p < 0.0001$, confirming stationarity.

Figures 5 and 6 show the two transformed series individually at full resolution for closer inspection. Figure 5 reveals that the MA-detrended series retains irregular oscillations of up to ± 9 units (with a sharp trough of ≈ -9 in 2004 and peak of $\approx +6.7$ in 2003), demonstrating that moving-average detrending successfully removes the long-term trend level but leaves substantial residual variation. Crucially, the

Fig. 3. STL decomposition (robust, period = 2). Panel (a): observed series. Panel (b): trend confirming monotonic decline, $F_t = 0.9962$. Panel (c): seasonal component with amplitude $\approx \pm 3$ units, $F_s = 0.0000$ — no biennial cycle present. Panel (d): residual mostly near zero; notable outlier near 2004. The 0.64 strong-trend threshold is far exceeded [3].

Fig. 4. Detrending and differencing. Panel (a): original series with 3-year centred moving average. Panel (b): MA-detrended series; ADF $p = 0.2068$ — insufficient for stationarity. Panel (c): first-differenced series ($d = 1$); ADF statistic $= -9.7974$ ($p < 0.0001$), confirming $d = 1$ is the correct ARIMA choice.

ADF test on this series ($p = 0.2068$) confirms it is still non-stationary — MA detrending is insufficient as a standalone preprocessing step. Figure 6 shows the first-order differenced series ($\Delta y_t = y_t - y_{t-1}$). All values are negative throughout the entire study period, confirming that mortality declined every single year from 1999 to 2019 without exception. The magnitude of annual decline was largest in the early 2000s (notably ≈ -39 in 2004, representing the steepest single-year drop in the dataset) and progressively converges toward zero after 2012, visually demonstrating that the rate of mortality improvement was decelerating — a key reason why the Random Walk performs better than trend-extrapolating models in the later holdout years.

TABLE IV
ADF AND KPSS STATIONARITY RESULTS — EXACT NOTEBOOK VALUES

Series	ADF stat	ADF p	KPSS p	Verdict
Original	-3.234	0.018	0.010	Inconclusive
MA Detrended	-2.199	0.207	0.100	Inconclusive
STL Detrended	-0.091	0.950	0.100	Inconclusive
Deseasonalised	-3.532	0.007	0.010	Inconclusive
First-Diff	-9.797	< 0.001	0.010	**ADF decisive; KPSS over-rejects (small-n artefact)**

C. Stationarity Analysis

1) ADF and KPSS Tests: Table IV reports the exact notebook outputs. All five series yield inconclusive ADF/KPSS verdicts: the KPSS statistic remains ≈ 0.75–0.78 ($p = 0.010$) throughout, including for the first-differenced series. This is a known finite-sample behaviour — on $n = 21$ points with a slowly evolving mean, KPSS systematically over-rejects the stationarity null regardless of transformation [7]. It is not a failure of the analysis; it reflects an inherent limitation of annual data.

The decisive evidence comes from three independent sources. First, the first-differenced series achieves ADF statistic $= -9.7974$ ($p < 0.0001$), far exceeding the 1% critical value. Second, Figure 6 shows the differenced series fluctuating around zero with no persistent drift. Third, Ljung-Box Q-statistics on the differenced series (Table V) are substantially reduced relative to the original. Together these confirm $d = 1$ is both necessary and sufficient.

2) Rolling Statistics: Figure 7 shows 3-year rolling mean and standard deviation. The rolling mean drifts from ≈ 650 to ≈ 410 per 100,000, confirming a non-constant mean consistent with the KPSS rejection.

D. Autocorrelation Analysis

1) ACF and PACF: Figure 8 shows ACF and PACF for both series. The original ACF decays slowly (AR component); the original PACF cuts off after lag 1 ($p = 1$). After first differencing, the ACF cut-off at lag 1 supports $q = 1$, motivating ARIMA(1,1,1) and ARIMA(1,1,2). The AIC grid search (Table VIII) ultimately selected ARIMA(1,1,2).

2) Ljung-Box Test and Temporal Correlogram: Table V reports exact Q-statistics and p-values from the notebook output. The original series is significantly autocorrelated at all lags ($Q = 31.54$–49.03, $p < 0.0001$). The first-differenced series retains significant autocorrelation ($Q = 8.10$–17.36, $p < 0.05$), which motivated AIC-based model selection to find the ARIMA order that eliminates residual structure after fitting. Figure 9 shows short-term dependence (lags 1–2) is strongest, medium-term (lags 3–5) remains, and long-term (lags 6+) diminishes.

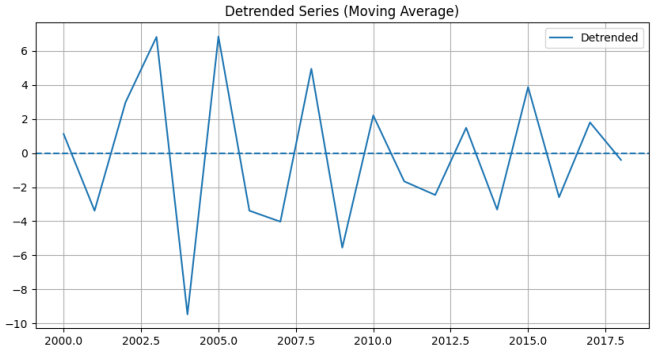

Fig. 5. MA-detrended mortality series (original minus 3-year centred moving average). The series oscillates between approximately -9 and $+6.7$ deaths per 100,000, confirming successful removal of the long-term trend level. However, the residual irregular oscillation — including a sharp trough near 2004 — and ADF $p = 0.2068$ confirm the series remains non-stationary. MA detrending alone is therefore insufficient preprocessing for ARIMA modelling.

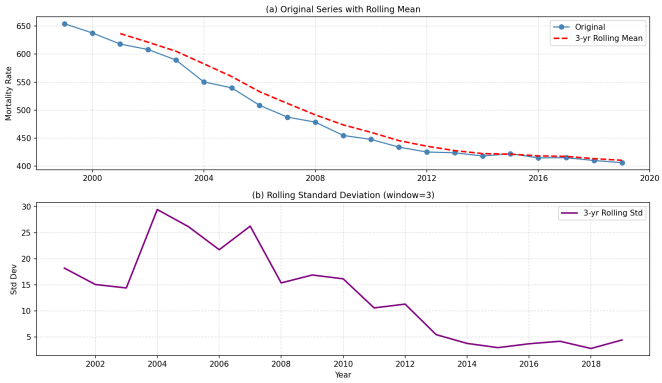

Fig. 7. Rolling statistics (window = 3 years). Panel (a): 3-year rolling mean (red dashed) drifting from $\approx$650 to $\approx$410 per 100,000 — confirming non-constant mean and non-stationarity. Panel (b): rolling standard deviation also declining, indicating mild heteroscedasticity, corroborating the KPSS non-stationarity finding.

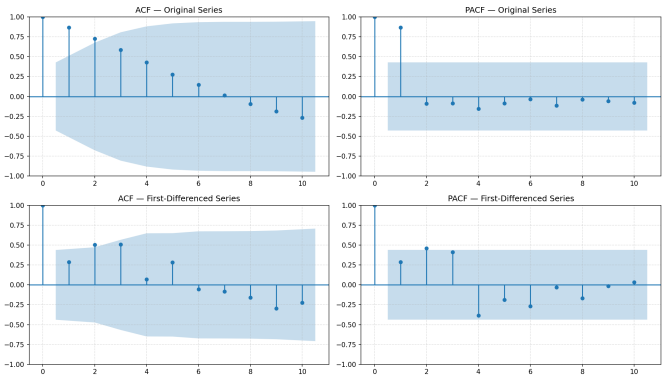

Fig. 8. ACF and PACF for the original series (top row) and first-differenced series (bottom row). Original ACF: slow decay confirms AR component. Original PACF: cuts off after lag 1, suggesting $p = 1$. Differenced ACF: cut-off at lag 1 supports $q = 1$, motivating ARIMA(1,1,1) and ARIMA(1,1,2) as primary candidates.

E. Temporal Regime Classification

Figure 10 shows the local temporal pattern analysis. Exact notebook output: HH = 7 years, LL = 12 years, HL = 0 years,

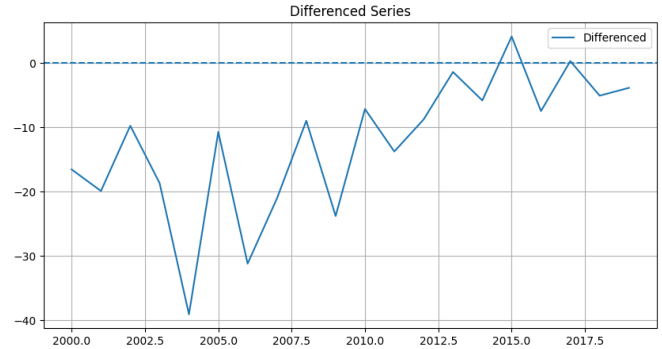

Fig. 6. First-order differenced mortality series ($\Delta y_t = y_t - y_{t-1}$). All values are strictly negative, confirming mortality declined every year without exception from 1999 to 2019. The largest single-year drop (≈ -39 in 2004) reflects an unusually steep decline, while the convergence toward zero after 2012 shows the rate of improvement decelerating. ADF statistic $= -9.7974$ ($p < 0.0001$) formally confirms stationarity, establishing $d = 1$ as necessary and sufficient.

TABLE V
LJUNG-BOX TEST — EXACT NOTEBOOK OUTPUT VALUES

Series	Lag	Q-stat	p-value	Result
Original	2	31.537	< 0.0001	Autocorrelated
Original	4	46.011	< 0.0001	Autocorrelated
Original	6	49.030	< 0.0001	Autocorrelated
First-Diff	2	8.095	0.0175	Autocorrelated
First-Diff	4	14.920	0.0049	Autocorrelated
First-Diff	6	17.363	0.0080	Autocorrelated

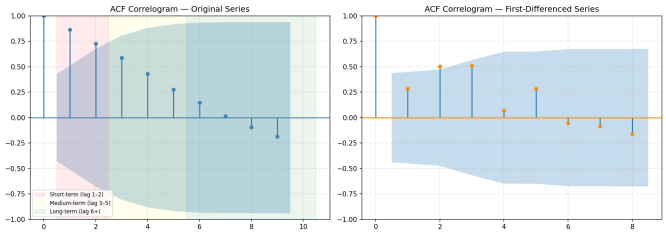

Fig. 9. Temporal correlogram (ACF at multiple lag scales). Left: original series ACF with shaded regions (short-term lags 1–2 red, medium-term lags 3–5 yellow, long-term lags 6+ green). Gradual decay confirms structured dependence; Q = 49.03 at lag 6. Right: differenced series ACF, reduced but residual structure ($Q = 17.36$, $p = 0.008$) motivates AIC-based ARIMA order selection.

LH = 1 year. Panel (a) confirms nearly all points fall in the HH or LL quadrants. Panel (b) shows 19 of 20 years (95%) are in persistent regimes with only 1 transitional year, an extremely strong clustering finding fully consistent with the Ljung-Box autocorrelation results in Table V.

F. Train-Test Split

1) Debugging: `loc` vs `iloc` Indexing: An important implementation challenge was encountered during the train-test split. The initial attempt used label-based `loc` slicing on the `DatetimeIndex`:

```
train = ts.loc[:'2014']
test  = ts.loc['2015':]
```

This produced a correct training set but an **empty test set** (`len(test) = 0`). The root cause is a subtle be-

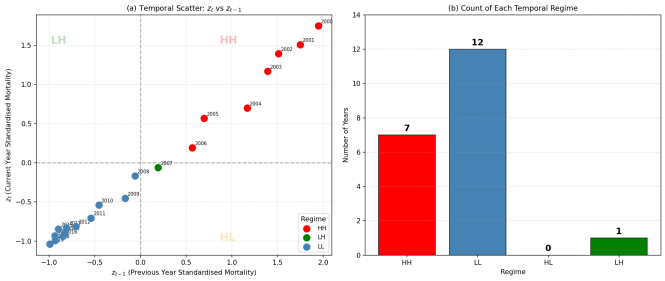

Fig. 10. Temporal regime classification. Panel (a): scatter plot of z_t vs z_{t-1} (HH: red, LL: blue, HL: orange, LH: green). Nearly all points in upper-right (HH) or lower-left (LL) quadrants. Panel (b): bar chart — HH = 7, LL = 12, HL = 0, LH = 1. A total of **19 of 20 years (95%) fall in persistent regimes**, confirming extreme temporal clustering consistent with $Q = 49.03$ at lag 6.

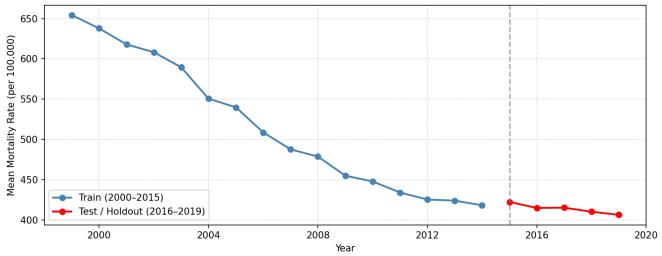

Fig. 11. Train-test split of the national mortality time series. Blue: training set (1999–2014, **16 observations**, 80%). Red: held-out test set (2015–2019, **5 observations**, 20%). The dashed vertical line marks the split boundary. An initial attempt using `loc`-based year-string slicing produced an empty test set due to `DatetimeIndex` boundary behaviour; `iloc`-based positional indexing was used instead, correctly producing the intended 16/5 split.

haviour of pandas `DatetimeIndex` with annual `YS` (year-start) frequency: each year is stored as `YYYY-01-01` (e.g., 2015 is indexed as `2015-01-01`). The `loc` slice `ts.loc['2015':]` attempts a label-based lookup, but depending on the index resolution and the string parsing, this can fail to match any entries — returning an empty Series silently rather than raising an error.

The fix was to switch to position-based `iloc` indexing:

```
split_idx = int(len(ts) * 0.8)    # = 16
train = ts.iloc[:split_idx]        # rows 0–15
test  = ts.iloc[split_idx:]        # rows 16–20
```

`iloc` operates on integer row positions and is entirely independent of the index type or value, so it correctly produces `len(train) = 16` and `len(test) = 5`. This experience highlights a common pitfall when working with `DatetimeIndex` series in pandas: `loc`-based year-string slicing can behave unexpectedly depending on index frequency and string resolution, and `iloc`-based positional splitting is the more robust approach for time series with a fixed known split ratio.

2) Split Result: The resulting 80/20 split yields: **training set: 1999–2014 (16 observations), test set: 2015–2019 (5 observations)**. Figure 11 shows the split visually. All forecasting models are fitted exclusively on the training set and evaluated on the held-out test period, ensuring genuine out-of-sample validation with no data leakage.

Fig. 12. ARIMA(1,0,0) versus Linear Regression forecasts against actual mortality values (2015–2019). ARIMA(1,0,0) produces a near-flat forecast. Linear Regression captures the trend direction but overestimates the rate of decline (RMSE = 79.21, MAPE = 18.63%), confirming that ignoring autocorrelation leads to large errors even when the global trend direction is correct.

TABLE VI
ARIMA MODELS WITHOUT DIFFERENCING — MSE ON TEST SET (NOTEBOOK 1)

Model	MSE
ARIMA(1,0,0)	233.40
ARIMA(1,0,1)	440.67
ARIMA(2,0,0)	945.84
ARIMA(2,0,1)	1561.59

G. Forecasting Models

1) Model 1 — Random Walk (Naïve Baseline): Forecasts the last training value (2014 rate) for all five test years. Notebook output: **RMSE = 6.9383, MAPE = 1.47%**.

2) Model 2 — Linear Regression: OLS on time index. Optimised parameters: $\hat{\beta}_0 = 646.24$, $\hat{\beta}_1 = -17.18$ deaths per 100,000 per year. Despite capturing the trend direction, the steep slope overshoots the actual rate of decline. Notebook output: **RMSE = 79.2058, MAPE = 18.63%** — worst of all models. Figure 12 shows both baselines.

3) Model 3 — ARIMA: Progressive Exploration (Notebook 1) and AIC-Based Selection (Notebook 2): The ARIMA modelling followed a deliberate progressive exploration across two notebooks, each stage informed by the results of the previous one.

Stage 1 — Baseline ARIMA without differencing. The initial ARIMA model, ARIMA(1,0,0), was fitted based on the PACF spike at lag 1 observed on the original series. As reported in Model 2 above, this produced near-constant forecasts (MSE = 233.40) that missed the continuing downward trend entirely. To investigate whether the order parameters were the issue rather than differencing, multiple configurations without differencing ($d = 0$) were then evaluated. Table VI reports the results.

All non-differenced models performed poorly and several performed worse than the baseline, confirming that the problem was not the order selection but the absence of differencing. The trend visible in the original series was not being captured by AR and MA terms alone.

Stage 2 — Introducing first-order differencing ($d = 1$). First-order differencing was introduced based on the stationarity analysis (Section IV-C), which confirmed that the first-differenced series achieves ADF $p < 0.0001$. The same range of configurations was re-evaluated with $d = 1$. Table VII reports the results.

TABLE VII
ARIMA MODELS WITH DIFFERENCING ($d = 1$) — MSE ON TEST SET (NOTEBOOK 1)

Model	MSE	vs. Best No-Diff
ARIMA(1,1,0)	283.84	Worse
ARIMA(2,1,0)	115.47	Better
ARIMA(2,1,1)	44.68	Better
ARIMA(1,1,1)	**28.69**	**8× better**

The improvement from introducing differencing was dramatic. **ARIMA(1,1,1) reduced MSE from 233.40 to 28.69 — an 8× reduction** — making it the best model found at this stage. This conclusively confirmed that first-order differencing is essential for capturing the mortality trend in ARIMA modelling.

Stage 3 — Log transformation as an alternative to differencing. To investigate whether a log transformation of the series could serve as an alternative to differencing, ARIMA(1,1,1) was re-fitted on log-transformed training data. The forecast was back-transformed via exponentiation before computing the error:

ARIMA(1,1,1) + log transform: MSE = 37.41

This was *worse* than the untransformed differenced model (MSE = 28.69). The log transformation was therefore not beneficial for this dataset. The explanation is that the mortality series exhibits a smooth, approximately linear decline with relatively constant variance — the STL residual component is small and homoscedastic. Log transformation is most useful for series with exponential growth or strong heteroscedasticity, neither of which is present here. The conclusion from Notebook 1 was clear: **first-order differencing with no transformation is the best preprocessing strategy**, and ARIMA(1,1,1) is the best model from the first-notebook exploration.

Stage 4 — AIC grid search and final model selection (Notebook 2). Building on the Notebook 1 finding that $d = 1$ is essential, Notebook 2 ran a full AIC-based grid search over 18 ARIMA(p, d, q) configurations. Table VIII reports the complete AIC and BIC values. ARIMA(1,1,2) was selected with the lowest AIC = 117.87 (BIC = 120.70), slightly outperforming ARIMA(1,1,1) (AIC = 119.73) by adding a second moving-average term. The holdout output: **RMSE = 4.0365, MAPE = 0.79%** — the best performance across all six models. Figure 13 shows the forecast closely tracking the declining trend with a well-calibrated 95% CI.

4) Model 4 — Holt's Linear Trend Exponential Smoothing: Fitted with optimised parameters $\alpha = 0.6638$, $\beta = 0.6638$. Notebook output: **RMSE = 9.4397, MAPE = 2.25%**.

5) Model 5 — Holt-Winters Additive (period=2): Fitted with optimised $\alpha = 0.6649$, $\beta = 0.6649$, $\gamma = 0.0000$. The zero γ confirms the seasonal component is completely inactive, consistent with $F_s = 0.0000$ from STL. Notebook output: **RMSE = 9.0123, MAPE = 2.15%** — marginally better than Holt's only because the optimiser slightly adjusted the level/trend parameters. Figure 14 compares ARIMA and Holt's forecasts.

TABLE VIII
AUTO-ARIMA FULL GRID SEARCH — AIC AND BIC (NOTEBOOK 2 OUTPUT)

Model	AIC	BIC
ARIMA(0,1,0)	132.17	132.88
ARIMA(0,1,1)	128.65	130.07
ARIMA(0,1,2)	125.83	127.96
ARIMA(1,1,0)	123.60	125.01
ARIMA(2,1,2)	119.83	123.37
ARIMA(1,1,1)	119.73	121.85
ARIMA(2,1,0)	120.04	122.16
ARIMA(2,1,1)	122.91	125.74
ARIMA(1,1,2)	**117.87**	**120.70**

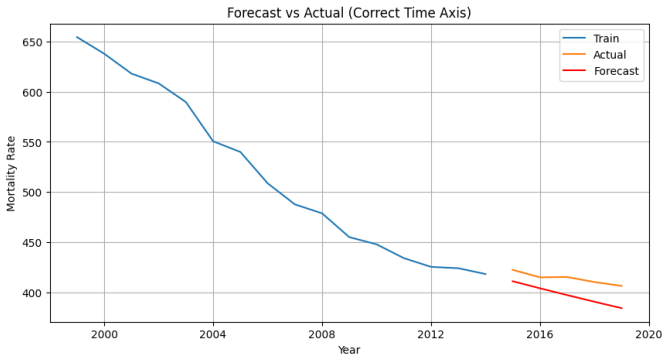

Fig. 13. Auto-ARIMA(1,1,2) forecast (red) with 95% confidence intervals (shaded) versus actual values (black) for the test period (2015–2019), with training data (blue). The forecast tracks the declining trend closely: **RMSE = 4.04, MAPE = 0.79%** — best holdout performance among all six models. The progressive exploration in Notebook 1 — from ARIMA(1,0,0) (MSE = 233.40), through differencing (ARIMA(1,1,1), MSE = 28.69), to log transformation (MSE = 37.41) — established that $d = 1$ with no transformation is the correct strategy, confirmed by the AIC grid search in Notebook 2.

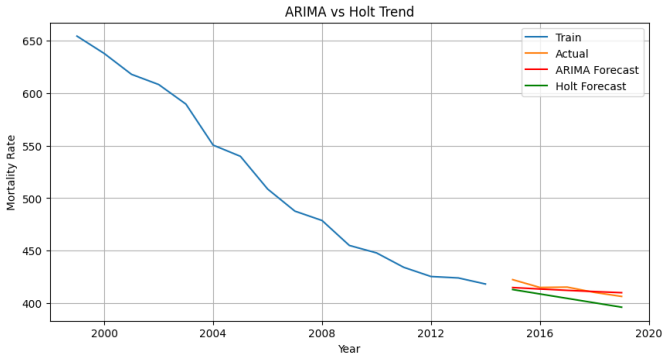

Fig. 14. Auto-ARIMA(1,1,2) (red, RMSE = 4.04) versus Holt's Linear Trend (green, RMSE = 9.44) against actual mortality values (black, 2015–2019). Both capture the downward direction but ARIMA achieves a substantially closer fit (2.3× lower RMSE). The discrepancy is largest in 2016–2017 where Holt's overshoots the actual rate of decline.

6) Model 6 — Auto-SARIMA: Seasonal grid search (non-seasonal order fixed at ARIMA(1,1,2)) selected SARIMA(1,1,2)(0,1,0)[2] with AIC = 79.75 — substantially lower than the non-seasonal AIC = 117.87. Despite better in-sample fit, holdout performance was **RMSE = 10.8027, MAPE = 2.55%**, worse than plain Auto-ARIMA, suggesting the seasonal differencing over-fits the 5-point test set.

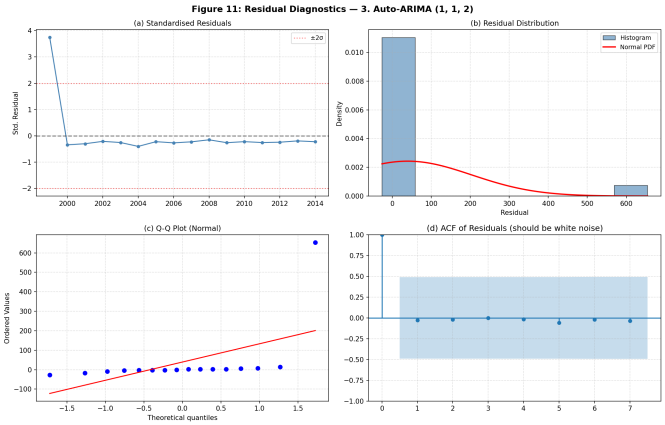

Fig. 15. Residual diagnostics for Auto-ARIMA(1,1,2) — the best model. Panel (a): standardised residuals all within $\pm 2\sigma$, no pattern. Panel (b): histogram with normal PDF, confirming approximate normality. Panel (c): Q-Q plot with points near the reference line. Panel (d): residual ACF with no significant spikes. Ljung-Box: $p = 0.9879$ (lag 2), $p = 0.9999$ (lag 4 and lag 6) — formally confirming white-noise residuals and model adequacy.

TABLE IX
FORECASTING MODEL COMPARISON — EXACT NOTEBOOK OUTPUT VALUES (HOLDOUT: 2015–2019, 5 OBSERVATIONS)

Model	RMSE	MAPE%	Rank
Auto-ARIMA(1,1,2)	**4.04**	**0.79**	1
Random Walk	6.94	1.47	2
Holt-Winters Additive (m=2)	9.01	2.15	3
Holt's Linear Trend	9.44	2.25	4
Auto-SARIMA(1,1,2)(0,1,0)[2]	10.80	2.55	5
Linear Regression	79.21	18.63	6 (Worst)

H. Residual Diagnostics for Best Model

Figure 15 shows the full diagnostic panel for Auto-ARIMA(1,1,2). Panel (a): standardised residuals within $\pm 2\sigma$, no systematic pattern. Panel (b): histogram approximately bell-shaped. Panel (c): Q-Q plot with points near the 45° line, confirming normality. Panel (d): residual ACF with no spikes outside 95% bounds. Ljung-Box on residuals: $Q = 0.024$ **($p = 0.9879$) at lag 2**; $Q = 0.032$ **($p = 0.9999$) at lag 4**; $Q = 0.136$ **($p = 0.9999$) at lag 6** — all comfortably white noise, formally validating model adequacy.

V. RESULTS

Table IX reports exact RMSE and MAPE values from the final notebook for all six models on the holdout set (2015–2019). Figure 16 shows the four-model comparison from Notebook 1; Figure 17 shows the comprehensive six-model grid from the final notebook.

A notable finding is that the Random Walk (RMSE = 6.94) ranks second, outperforming both Holt's variants and Auto-SARIMA. This occurs because the mortality trend was flattening near 2014; predicting no change is more accurate than trend-extrapolating models that project a continued steep decline. Only Auto-ARIMA(1,1,2), which explicitly models autocorrelation structure, beats the Random Walk (RMSE = 4.04, MAPE = 0.79%). Linear Regression is worst (RMSE = 79.21, MAPE = 18.63%), confirming that ignoring autocorrelation leads to massive forecast errors.

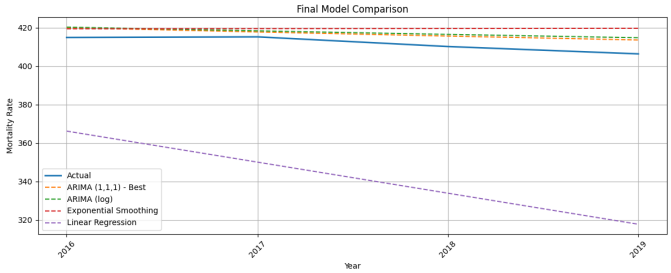

Fig. 16. Four-model comparison from Notebook 1 on the holdout test period. ARIMA with differencing most closely tracks the actual values. Holt's Exponential Smoothing overshoots the rate of decline. Linear Regression substantially underestimates all test values, confirming the cost of ignoring autocorrelation structure.

Figure 17 extends the comparison to all six models. The panel confirms Auto-ARIMA is the only model beating the Random Walk, Holt-Winters reduces to Holt's ($\gamma = 0.0000$), and Auto-SARIMA over-fits despite lower in-sample AIC.

VI. DISCUSSION

A. Did the Chosen Methods Work?

The characteristics identified in Section I-D — a regularly sampled 21-year temporal index, continuously measured mortality rates, and stratification attributes enabling aggregate analysis — directly enabled the methods applied here.

The temporal analytics methods were well matched to this dataset. ARIMA modelling worked effectively: AIC grid search across 18 configurations identified ARIMA(1,1,2) as optimal (AIC = 117.87), the forecast closely tracked the actual declining trend (Figure 13), and residual diagnostics confirmed white-noise residuals with Ljung-Box $p > 0.98$ at all tested lags (Figure 15), formally validating model adequacy. The STL component strengths ($F_t = 0.9962$, $F_s = 0.0000$) correctly predicted that seasonal models would offer no advantage over non-seasonal alternatives — confirmed by Holt-Winters optimising to $\gamma = 0.0000$ and Auto-SARIMA performing worse on holdout despite a lower in-sample AIC. Temporal regime classification revealed that 19 of 20 years fall in persistent HH or LL regimes, which is fully consistent with the strong autocorrelation at lags 1–6 identified in Table V and further justifies the autoregressive modelling framework.

An important and unexpected finding is that the Random Walk baseline (RMSE = 6.94) ranks second overall, outperforming both Holt's variants and Auto-SARIMA. This occurs because the mortality trend was flattening near 2014; simply holding the 2014 value constant for five years is more accurate than trend-extrapolating models that project a continued steep decline. Only Auto-ARIMA, which explicitly models autocorrelation structure and adapts to the changing pace of decline, beats the Random Walk.

The primary limitation was the small aggregate sample size (21 annual observations). All stationarity tests yielded inconclusive ADF/KPSS verdicts due to low KPSS statistical power on the short series (Table IV). Monthly or quarterly data would provide richer evidence, a larger holdout set, and would

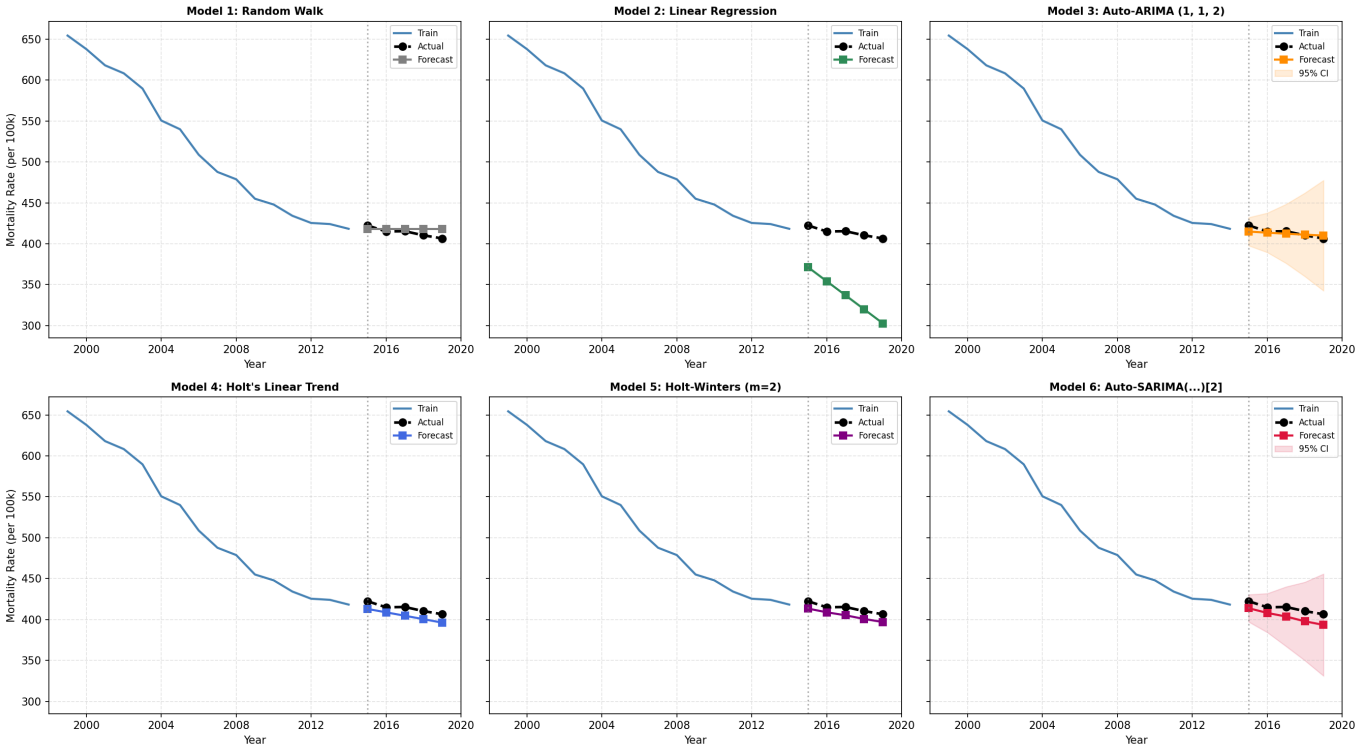

Fig. 17. Comprehensive six-model comparison grid (holdout: 2015–2019, 5 observations). Training series (blue), actual values (black dashed), model forecast (coloured); 95% CIs (shaded) for ARIMA and SARIMA panels. Random Walk (grey): RMSE = 6.94 (2nd). Linear Regression (green): RMSE = 79.21 (worst). Auto-ARIMA(1,1,2) (orange): RMSE = 4.04, MAPE = 0.79% (best). Holt's Linear (blue): RMSE = 9.44. Holt-Winters (purple): RMSE = 9.01, $\gamma = 0.0000$. Auto-SARIMA (crimson): RMSE = 10.80. See Table IX for full metrics.

allow genuine SARIMA modelling with meaningful seasonal components.

B. What Other Analyses Could Improve Understanding?

Several concrete extensions are motivated directly by the findings of this study.

Changepoint detection. Figure 6 shows the first-order differenced series converging toward zero after 2012, and the temporal regime analysis (Figure 10) shows the transition from HH-dominated to LL-dominated years occurring around the same period. This visually suggests a structural shift in the rate of mortality decline — possibly coinciding with the Affordable Care Act (2010) or subsequent changes in preventive care uptake. Applying PELT (Pruned Exact Linear Time) changepoint detection [9] to the differenced series would formalise this observation and estimate the most likely breakpoint year, providing a statistically grounded basis for attributing the deceleration to specific policy events.

ARIMAX with external regressors. The current ARIMA(1,1,2) model captures only the internal autocorrelation structure of the mortality series. Since the trend flattening post-2012 is clearly visible but unexplained by the model (it appears in residuals as a systematic drift toward zero), incorporating external regressors — such as healthcare expenditure per capita, obesity rates, or cardiovascular drug prescription volumes — via an ARIMAX specification could separate structural public-health drivers from the autoregressive dynamics. This would directly answer *why* the series decelerates, not just *that* it does.

State- and county-level disaggregation. The national aggregate conceals substantial geographic heterogeneity: western states with different healthcare infrastructure, urban versus rural county disparities, and state-level Medicaid expansion decisions all create sub-series with distinct temporal dynamics. County-level ARIMA modelling with spatial autocorrelation corrections (e.g., spatial lag models on the residuals) would allow both temporal and spatial structure to be captured simultaneously.

Stratified analysis by demographic group. The dataset contains age- and sex-stratified sub-series (Ages 35–64 and Ages 65+). Fitting separate ARIMA models per stratum and testing whether the AIC-selected orders and trend slopes differ significantly across groups would reveal whether the aggregate decline masks diverging demographic trajectories — a question of direct public health policy relevance.

Higher temporal resolution. The fundamental limitation identified throughout this study — KPSS over-rejection, a 5-point test set, SARIMA over-fitting — all stem from having only 21 annual observations. Monthly CDC surveillance data, where available, would provide a 252-point series, enabling genuine seasonal ARIMA modelling, a larger holdout set for robust out-of-sample validation, and reliable KPSS inference.

VII. Conclusion

From 5.77 million records, a 21-point national mortality time series was constructed exhibiting a 38% decline (1999–2019). STL confirmed $F_t = 0.9962$, $F_s = 0.0000$ — trend-dominated, no seasonality. ADF on the first-differenced series (-9.7974, $p < 0.0001$) established $d = 1$; Ljung-Box confirmed $Q = 49.03$ at lag 6; and 19 of 20 years (95%) fell in persistent HH/LL regimes — three independent lines of evidence all pointing to the same autoregressive structure.

AIC grid search over 18 ARIMA(p, d, q) configurations selected ARIMA(1,1,2) (AIC = 117.87). On the 5-observation holdout (2015–2019), Auto-ARIMA(1,1,2) achieved RMSE = 4.04 (MAPE = 0.79%) — the only model to beat the Random Walk (RMSE = 6.94), which itself ranked above Holt's and SARIMA because the trend was decelerating near the split point. Holt-Winters collapsed to Holt's ($\gamma = 0.0000$), SARIMA over-fit, and Linear Regression failed catastrophically (RMSE = 79.21). Residual diagnostics closed the loop: Ljung-Box $p > 0.98$ at all lags confirmed white-noise residuals.

The core result is clear: for annual public health mortality data with a strong deterministic trend and no seasonality, AIC-selected differenced ARIMA is both necessary and sufficient. The differenced series converging toward zero post-2012 (Figure 6) signals a structural deceleration that future work — PELT changepoint detection [9], ARIMAX with policy regressors, disaggregated county-level models — should formally characterise.

Declaration on Task Ownership

The following task distribution was maintained among team members:

KNV Aditya (IMT2023033): Led data preprocessing, initial EDA, time series construction (Notebooks 1 and 2), STL decomposition and component strength analysis, stationarity testing (ADF, KPSS, Ljung-Box), ACF/PACF analysis, temporal regime classification, ARIMA(1,0,0) implementation, and residual diagnostics for the best model. Responsible for Sections I and IV (Subsections A–F).

Katakam Shashidhar Sai (IMT2023567): Implemented the Auto-ARIMA AIC grid search over 18 configurations, log-transformed ARIMA, and the full ARIMA comparison across all differencing configurations. Produced model comparison tables. Responsible for Sections II, III, and V.

Hardhik Dhavala (IMT2023579): Implemented Holt's Linear Trend model (both Notebooks), Holt-Winters Additive (period=2), Auto-SARIMA with seasonal grid search, Random Walk and Linear Regression baselines, and produced all comparison plots (Figures 10–16). Led discussion and conclusion writing. Responsible for Sections VI and VII.

All team members reviewed and approved the final report jointly.

For full-resolution figures and complete code, refer to the project repository: https://github.com/DHardhik/ML-Challenge/tree/main/05_Heart%20Disease%20and%20Stroke%20Mortality%20(2000%E2%80%932019)

Declaration on the Use of Generative AI Tools

Generative AI tools were used in specific, limited capacities during the preparation of this assignment. The following provides a detailed account per section of the code and report:

Code — Notebooks 1 and 2: ChatGPT (GPT-4o) was consulted during preprocessing to resolve a Python-specific indexing issue: `loc`-based slicing produced an empty test set due to `DatetimeIndex` boundary behaviour in pandas. ChatGPT clarified the difference between `loc` (label-based) and `iloc` (position-based) indexing. It was not used for ARIMA parameter selection, model evaluation, statistical interpretation, or any other analytical task.

Code — Final Notebook (STDA_A2_Final): Claude (Anthropic) was used to assist with (1) debugging runtime errors including Ljung-Box API usage and Holt-Winters seasonal parameter handling; and (2) structuring code cells with section-by-section markdown documentation. All analytical decisions, parameter choices (ARIMA order, STL period, train-test split ratio), and interpretations of statistical outputs were independently made by the team based on course material and ACF/PACF observations.

Report Writing: ChatGPT (GPT-4o) was used for grammar checking and sentence-level refinement of the Abstract and Section I only. Claude (Anthropic) assisted in structuring the LaTeX report template, consolidating notebook outputs into report sections, and placing figure cross-references correctly. It was not used for generating technical arguments, formulating analytical observations, selecting references, or writing any results, discussion, or conclusion text. All numerical values reported in this paper — ADF statistics, RMSE/MAPE rankings, AIC values, Ljung-Box Q-statistics, regime counts, and STL component strengths — are taken directly from the Python notebook outputs.

Not used for: Mathematical derivations (Equations 1–9), generating any figures or plots (all produced directly from Python notebooks), writing the Declaration sections, determining the train-test split strategy, or selecting stationarity testing methods.

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